



sea pens 103
 sea slugs 77, 121
 sea snails 120, 121, 139, 173, 177, 183
 sea snakes 184, 185
 sea squirts 138, 156
 sea stacks 150, 151, 153, 166
 sea urchins 113, 115, 128
 seagrasses 133, 168, 169, 182–183
 seahorses 183
 seals 78, 79, 111, 155, 190, 193, 194, 198,
 199, 200, 208–211, 212, 213
 dangers to 238
 seamounts 17, 28, 29, 40, 41, 65
 seashells 46
 seawater 22, 48–49
 color of 50, 72
 freezing point of 193
 seaweeds 62, 72, 73, 78, 110, 112–115, 156,
 162, 168, 183, 185
 giant kelp 114–115, 117
 sediments 29, 46, 65, 110, 129, 174,
 175, 176
 shadow zones 71
 shale 46
 sharks 64, 71, 79, 86–87
 filter-feeding sharks 88–89, 109
 reef sharks 137

 shellfish 104, 154, 161, 232, 241
 as prey 139, 162, 163, 169, 173, 181
 shingle 44, 149
Shinkai 6500 225
 shipping 191
 ships 18, 39, 51, 57, 135, 191, 203, 218,
 219, 236–237
 cruise ships 237
 fishing 232–233
 research ships 219, 220
 shipwrecks 45, 110, 111, 221, 225,
 226–227
 shrimp 51, 71, 104, 139, 157
 as prey 96, 163, 168, 181
 silt 45
 sliding boundaries 31
 sonar 221
 sounds 51
 Southern Ocean 9, 18–19, 51, 153, 188,
 200–201
 animals 74, 75, 91, 96, 97, 194–195,
 198–201
 currents 67
 waves 57
 spermaceti 95
 spits 159
 sponges 138, 139
 spoonbills 163
 squid 71, 98, 99, 122, 195
 giant squid 101
 as prey 94, 96, 97, 111
 starfish 128–129, 139
 stargazers 118
 stilts 163
 stingrays 110
 stings 76, 77, 130, 131

storm surges 55
 storms 45, 54–55, 109, 149, 241
 streamlined bodies 80, 82, 122, 168, 197
 subduction zones 12, 32, 33, 34, 35, 36
 submarines 10
 submersibles 33, 220, 224–225, 227
 summer 10, 188, 189
 Sun 22, 48, 52, 147, 188
 sunlight 50–51, 70, 72, 78, 108, 110, 112,
 114, 132, 188
 sunlit zone 67, 70, 71, 72–77, 108
 surfing 59, 158, 172
 surgeonfish 136
 surveying 221
 swordfish 83
 symbiosis 132

T
 temperature 10, 50, 104, 105
 tentacles 76, 77, 98, 103, 111, 122, 123,
 129, 130, 131, 132, 156, 161
 thermocline 71
 tidal bores 172
 tidal power 231
 tidal races 147
 tides 113, 146–147, 156, 157, 172, 176,
 177, 178
 high tide 38, 147, 161
 low tide 160
 neap tide 147, 176
 spring tide 147, 176
Titanic 202, 224, 227
 tourism 158, 202, 239, 243
 trade 18, 218, 219, 227, 236–237
 trenches 12, 15, 28, 29, 32–33
Trieste 33
 tripodfish 103
 tropical oceans 15, 71, 178–179
 animals of 96, 97, 131, 132, 136–139,
 169, 171
 currents 61
 temperature 50
 winds 50, 55
 tsunamis 15, 38–39
 tubercles 91
 tuna 82, 83
 turbines 230
 turnicates 138
 turnstones 162
 turtles 63, 133, 168–169, 175, 182–183,
 184, 244
 dangers to 238, 242
 tusks 95, 208
 twilight zone 70, 71, 75, 98–99, 224
 typhoons 55

U
 upwelling 64–65

V
 venom 123, 131, 137, 139, 161, 185
Vasa 226
 Vikings 218
 viperfish 99
 volcanoes 12, 17, 25, 26–27, 29, 30, 34–35,
 38, 40–43, 142, 200, 201

W
 walruses 208
 Walsh, Don 33
 water cycle 48
 water, sea 48–49, 222
 color of 50, 72
 freezing point of 193
 waves 38–39, 56–59
 energy from 231
 erosion 44, 148–149
 weevs 161
 whales 51, 67, 71, 79, 127, 135, 201
 baleen whales 90–93
 beluga whales 189, 221
 blue whales 91
 bowhead whales 90
 Bryde's whales 79
 gray whales 90
 humpback whales 91, 93
 minke whales 91
 narwhals 95
 pygmy right whales 91
 sperm whales 94, 95, 101
 toothed whales 94–95
 whaling 201
 whirlpools 147
 whiskers 208
 wind 15, 18, 50–51, 52, 65, 158, 159
 and currents 60
 and trade 236
 and waves 56, 57
 wind power 230
 winter 10, 18, 50, 61, 71, 188, 189, 195
 worms 104, 105, 111, 160, 161

Z
 zones of the ocean 70–71
 zooplankton 74–75, 78, 192

